

AYUSH JUVEKAR

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ML / Data Scientist

PROFILE

Computer science graduate with applied ML and data-analysis experience across model evaluation, computer vision, and privacy/security-focused Python workflows.

EDUCATION

Michigan Technological University, Houghton, MI
M.S. Computer Science

CGPA: 3.9/4.0
Graduated Apr 24, 2026

Pune Institute of Computer Technology, Pune, India
B.E. Computer Engineering

GPA: 8.14/10
Jun 2024

EXPERIENCE

Freelance Software Developer | Coherence Academy
Pune, India

Aug 2026 - Present

- Building Coherence Fee Desk, a Next.js platform replacing the academy's spreadsheet-based admissions, fee collection, and reporting.
- Separately run the software for Coherence-backed events (Campus Connect, Mission 95): registration, UPI payments, Sheets sync.

Teaching Assistant - Physics Department | Michigan Technological University
Houghton, MI

Sep 2024 - Apr 24, 2026

- Supervised and graded physics labs for 100+ students across four semesters, guiding experiments, explaining concepts, and providing clear feedback across five lab sections per semester.

Technical Intern | CVBDSL Lab, SCTR's Pune Institute of Computer Technology
Pune, India

Feb 2023 - May 2023

- Analyzed RSA and related encryption approaches in Python/Jupyter during a four-month academic internship, comparing implementation tradeoffs for computational cost and efficiency.

SELECTED PROJECTS

Crowd Anomaly Detection Prototype | *Python, TensorFlow, OpenCV, TensorFlow Lite, Jetson Nano*

- Led model research and data collection for a BE final-year team project on edge crowd-anomaly detection, recording training videos and evaluating CNN/TensorFlow approaches on 10K+ frames.
- Optimized the prototype with TensorFlow Lite for Jetson Nano demonstration, reaching about 92% test accuracy in controlled demo runs.

Particle Vibration Analysis | *MATLAB, Computer Vision, Object Detection, ZED 2 Camera*

- Built MATLAB computer-vision scripts for particle vibration analysis from ZED 2 camera video, using object detection workflows to track particles in test recordings.
- Delivered analysis outputs for a professor's downstream PhD research, validating scripts on sample particle videos before larger research use.

TECHNICAL SKILLS

Languages Python, SQL, C++, TypeScript

ML / Data TensorFlow, Scikit-learn, NumPy, Pandas

Computer Vision OpenCV, image/video preprocessing

Experimentation Model evaluation, dataset preparation, feature analysis

Tools Jupyter Notebook, Git, Docker

Databases PostgreSQL, MongoDB

PUBLICATION

A. Ghadge, A. Juvekar et al., "Carbon Monoxide Concentration Monitoring System for Automating Air Filters," *ESCI 2023*. [IEEE](#)